IMMERSIONS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3564-3576, following the statement at 3546-3562.

Editorial note. The third part 'a combination of the first two' factors an immersion as closed-after-open and applies (1) and (2).

Allowed standard prerequisites (editorial).

- Restricting maps to a closed subspace. lemma-restrict-map-to-closed (same chapter, cited at ~3567). The closed subspace cut by Im(g^*I -> O_Y) is the LRS fibre product.
- Closed subspaces of schemes are schemes. lemma-closed-subspace-scheme (card L26399, cited at ~3571). Z is a scheme, hence equals the scheme fibre product.
- Open fibre products of affines. lemma-open-fibre-product (same chapter, cited at ~3573). Engine for the open immersion case.

Source reference. Original lemma 01IN: sources/stacks-schemes/pinned-original-source.tex, source lines 1961–1961. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01JR: sources/stacks-schemes/pinned-original-source.tex, source lines 3415–3415. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HQ: sources/stacks-schemes/pinned-original-source.tex, source lines 673–673. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ and $g : Y \to S$ be morphisms of schemes with the same target.*

(1) *If $f : X \to S$ is a closed immersion, then $X \times_S Y \to Y$ is a closed immersion. Moreover, if $X \to S$ corresponds to the quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_S$, then $X \times_S Y \to Y$ corresponds to the sheaf of ideals $\mathrm{Im}(g^*\mathcal{I} \to \mathcal{O}_Y)$.*

(2) *If $f : X \to S$ is an open immersion, then $X \times_S Y \to Y$ is an open immersion.*

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(3) *If $f : X \to S$ is an immersion, then $X \times_S Y \to Y$ is an immersion.*

Proof. Assume that $X \to S$ is a closed immersion corresponding to the quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_S$. By Lemma 01HQ the closed subspace $Z \subset Y$ defined by the sheaf of ideals $\mathrm{Im}(g^*\mathcal{I} \to \mathcal{O}_Y)$ is the fibre product in the category of locally ringed spaces. By Lemma 01IN Z is a scheme. Hence $Z = X \times_S Y$ and the first statement follows. The second follows from Lemma 01JR for example. The third is a combination of the first two. □